

Open System-One Decision Models: Frozen Diffusion-LM Backbones and Specialist Decision Heads for Calibrated Scientific Decisions

Anonymous

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Abstract

Structured decisions—selecting among options, verifying whether evidence supports a claim, and grading response quality on ordinal scales—are the atomic operations of automated scientific pipelines. Proprietary “System-One” models deliver such decisions as typed API calls without generation, but are closed. We present an open, trainable counterpart: a frozen masked-diffusion language model backbone (LLaDA-8B) with small specialist decision heads producing probabilities for three decision types (choice, noul, score). We contribute three findings. (1) A deterministic canonical ordering gives a permutation-invariant input sequence for options with distinct token representations, rather than relying on train-time order augmentation. (2) Multi-layer attention pooling over option spans outperforms last-layer mean pooling for choice (+7 points), and the combined specialist, canonical, and ordinal recipe improves graded scoring (+9 points). (3) Scaling training data 30× on an out-of-domain corpus improves in-distribution accuracy to 0.96 but does not transfer (−13 points on the target domain), suggesting that volume alone is insufficient. On public benchmarks, SCIEV reaches 0.8500 on SciFact claim verification and transfers zero-shot to general classification (0.9300 SST-2, 0.8537 AG News); knowledge-intensive GPQA remains difficult (0.3125 on main). Candidates using LoRA domain adaptation score higher on elite choice but lower on GPQA and SciFact. Their head-training schedules also differ, so this is a candidate comparison, not an isolated causal test of DAPT. The release retains the frozen-backbone heads.

1 Introduction

Automated scientific pipelines—agentic literature review, claim verification, data extraction, routing to slower reasoning components—are built out of small structured judgments: *which* of these options answers the question, *does* the evidence support the claim, *how* good is this candidate answer. Today these judgments are typically extracted from generative LLMs by prompting for JSON and parsing the output, a pipeline that is slow, brittle under format drift, and poorly calibrated.

A recent class of models removes generation entirely: given an unstructured state and a set of typed questions, a *System-One* model returns typed values—a choice among options, a yes/no probability, an ordinal score—each with a calibrated probability, in a single forward pass. The flagship proprietary model of this class is Jev (TypeSafe AI, 2026); open reimplementations appeared within days of its release (kev, openjev, laya, SemIf), all built on autoregressive or encoder backbones.

We ask what an open System-One model needs to be good at, and build one on the other major family of language models: a *masked-diffusion* backbone (LLaDA-8B), whose bidirectional attention is a natural fit for decision readouts that compare each option against the full context.

To preserve the pretrained weights and limit trainable parameters, we train specialist heads per decision type on frozen multi-layer hidden states. Our contributions:

- **Permutation invariance by canonical ordering.** Sorting options deterministically by content makes the presentation order a function of the option set. For distinct token representations, permutations produce the same input sequence; train-time order averaging alone does not impose this constraint.
- **Specialist heads per decision type.** Multi-layer attention pooling over option spans improves choice (+7 points over mean pooling) while hurting binary verification—different decision types want different readouts. The combined specialist, canonical, and ordinal recipe improves graded scoring (+9 points).
- **Data volume and adaptation are not sufficient.** A $30\times$ out-of-domain data increase fails to transfer (-13 points). We also compare release candidates with and without LoRA domain adaptation. The adapted candidates improve some elite metrics but lose accuracy on GPQA and SciFact. We retain the frozen-backbone candidate; matching the head optimization schedules is necessary before drawing causal conclusions about adaptation.

We evaluate on the elite scientific battery, GPQA, SciFact, and four classification datasets. Metrics cover accuracy, calibration, permutation consistency, and retrospective automation at 5% error.

2 Setup

Decision types. We consider three typed decisions over a state (context) and a question: **choice** selects one of K answer options; **noul** decides whether evidence supports a claim ($K=2$); **score** assigns an ordinal grade $0 < 1 < 2$ (unrelated / related-but-wrong / correct).

Model. The released span-pooling heads encode context, question, and options as one observed token sequence, without inserting mask markers or generating text. A frozen masked-diffusion backbone (LLaDA-8B-Instruct) produces token representations. A mean-pooled MLP reads final-layer option spans for noul and score; *AttnPool* learns a query for each of layers $\{8, 16, 24, 32\}$ and concatenates the pooled features for choice. The released choice head has 67,166,209 parameters; each MLP has 16,793,601. Only these heads are trained during decision training.

Canonical ordering. Options are sorted by token-id content before encoding, and gold indices remapped. For options that remain distinct after tokenization and truncation, any input permutation collapses to the same sequence. Identical or truncated-to-identical options require a separate tie policy if their external identifiers must remain distinguishable.

Ordinal auxiliary loss. For score we add a CORAL-style term $\sum_{j=1}^{K-1} \text{BCE}(P(y \geq j), \mathbf{1}[g \geq j])$ on cumulative softmax mass, respecting the grade order without changing the K -way decode.

Metrics. We report accuracy, expected calibration error (ECE), option-order flip rate (a change in the selected option’s original identity after permutation), and *automation@5%err*: the largest confidence-ranked prefix with realized error $\leq 5\%$. Automation is retrospective on labeled evaluation data, not a guarantee of deployment error. Historical non-canonical flip figures compared

output positions without remapping option identities; they are excluded here pending re-evaluation. The release fixes that metric and includes regression tests.

3 Experiments

3.1 Scientific decision battery

The *elite* battery is built from passage-grounded scientific QA typed by reasoning kind (factual, causal, multihop, negation, numerical, definitional); each passage generates a choice item ($K=4$, one gold + three distractors), a noul item (supported vs. unsupported claim), and a score item (correct / related-but-wrong / unrelated answer). Splits are passage-disjoint to prevent leakage (2,240/6,720/6,720 train rows per type). The *v2* battery scales the same construction to 25,000 passages from three raw QA corpora (physics/polymer LaTeX full texts, astronomy, and a derived set) with groundedness filters—content-word recall ≥ 0.5 and numeric binding—yielding $\sim 470,000$ training decisions; we use it to probe the data-scaling question.

3.2 Public benchmarks

GPQA main (448 graduate-level questions) and diamond subset (198) as choice; SciFact dev (340 claim–abstract pairs) as noul and 3-way score; plus general classification (SST-2, AG News, Enron spam, Banking77) to probe out-of-domain transfer. Benchmark examples are not used for gradient updates or temperature fitting; temperatures are fitted on the battery dev split. Benchmark outcomes did inform the choice of release candidate.

3.3 Candidate protocols and comparison limits

The frozen release heads (**c_***) use learning rate 3×10^{-4} , accumulation 1, warmup 200, and 2,000/3,000/3,000 example steps for choice/noul/score. Heads for the adapted candidates (**dag_***, **da_***) and the small-backbone control (**eb_***) instead use learning rate 10^{-3} , accumulation 8, warmup 100, and 3,000 steps per type. All use seed 7331, the same elite training files, the same head architecture per type, and an ordinal auxiliary weight of 1 for score. Thus these are candidate comparisons, not a controlled DAPT-only ablation.

DAPT uses LoRA rank 16, alpha 32, dropout 0.05, and the masked-token objective implemented in `sciev.dapt`. With batch size 16 and sequence cap 1,024, the 2,000/5,000-step adapters correspond to at most 32.8/81.9 million input token slots, not measured counts of unique non-padding tokens. The 5,000-step run warm-starts from the 2,000-step adapter with a fresh optimizer and a schedule recomputed for the shorter total budget; optimizer and RNG state were not restored. The 1.4B **bw1_sr** control is checkpoint-g20 from a 20-step run, with a 768-token window versus 2,048 for LLaDA. It is an early-training control, not evidence about the capacity of a fully trained 1.4B model.

4 Results

4.1 Ablations on the scientific battery

Key findings: train-time logit averaging costs choice accuracy in these runs, while canonical ordering deterministically fixes presentation order. AttnPool on intermediate layers helps choice (+7) but hurts noul, motivating per-type heads. The final combined recipe lifts score from 0.522 to 0.608; this comparison alone does not isolate the ordinal loss from the other changes.

	choice	noul	score
Baseline (mixed MLP)	0.775	0.707	0.522
+ orders= 4	0.721	0.782	0.586
+ AttnPool	0.847	0.636	0.588
+ both	0.749	0.780	0.576
Per-type heads	0.876	0.710	0.574
Per-type + canon. + ord.	0.870	0.783	0.608

Table 1: Accuracy ablations on the scientific battery (frozen LLaDA-8B). The release combines specialist heads, canonical ordering, and ordinal scoring. Historical, unaligned positional flip figures are not reported.

4.2 Data scaling does not transfer across domains

	in-distribution (v2)	elite (target)
choice	0.965	0.738
noul	0.942	0.782
score	–	0.616

Table 2: Heads retrained on $30\times$ more data from a different QA distribution (qa_raw corpora) reach near-ceiling in-distribution accuracy but do not transfer to the elite battery, whose questions stress causal/multihop/negation reasoning absent from the source corpus.

4.3 Public benchmarks

4.4 Backbone and adaptation candidates

Relative to c_* , the g5000 candidate gains 1.38 percentage points on elite choice but loses 5.36 on GPQA main and 12.65 on SciFact noul. The g2000 candidate improves retrospective elite noul automation from 0.2777 to 0.4125. Neither meets the release gate: at most one point of elite accuracy loss per type and either improved SciFact or at least three points of GPQA-main improvement. These observations support retaining c_* among the tested candidates, not rejecting DAPT in general.

4.5 Matched frozen/adapted comparison on complete evidence

The candidate screen above confounds adaptation with head optimization and uses silently truncated contexts. The `systemone-v2` battery removes both problems: identical recipe `scientific-v1` (spanpool, canonical order, per-type heads, 3×10^{-4} head LR, warmup 200, accumulation 1, seed 7331), group-disjoint splits, and evidence that either fits whole or is excluded with a recorded reason (8,576 long-context exclusions; median passage 783 tokens).

Calibration (dev-fitted temperature, frozen at evaluation) is strong in-distribution under seed 7331: elite ECE is 0.0167/0.0189/0.0118 for choice/noul/score under the frozen arm, versus 0.0164, 0.0163, and 0.0823 under the adapted arm. On SciFact noul the frozen arm sits below its own majority baseline (0.483 mean versus 0.5964), while the adapted arm reaches 0.709 mean accuracy (macro-F1 0.7289, ECE 0.1607 at seed 7331): domain adaptation transfers to an external verification regime it was never trained on. The cost concentrates in rubric scoring, where adaptation

benchmark	type	n	accuracy	ECE	auto@5%
GPQA main	choice	448	0.3125	0.3180	0.0112
GPQA diamond	choice	198	0.3182	0.3089	0.0152
SciFact dev	noul	340	0.8500	0.0998	0.2706
SciFact dev	score	340	0.6294	0.0796	0.2559
Enron spam	choice	2000	0.7510	0.1046	0.0000
SST-2	choice	872	0.9300	0.0682	0.9415
AG News	choice	7600	0.8537	0.0556	0.3674
Banking77 ($K=77$)	choice	3080	0.2289	0.0809	0.0195

Table 3: Public-benchmark results for the frozen `c_*` release heads, transcribed from the archived evaluation JSONs rather than earlier rounded summaries. Benchmark data were not used for gradient updates or temperature fitting. Temperatures fitted on elite dev were 0.8274, 2.7127, and 0.9367 for choice, noul, and score. Counts and metrics describe our converted evaluation subsets; third-party results with unverified protocol equivalence are not included as directly comparable rows. Machine-readable values and source hashes are in `paper/results.json`.

candidate	elite			GPQA		SciFact	
	choice	noul	score	main	diamond	noul	score
bw1_sr 1.4B (g20)	0.2802	0.6672	0.3344	0.2656	0.2828	0.5941	0.3000
LLaDA-8B frozen (c_*)	0.8698	0.7825	0.6075	0.3125	0.3182	0.8500	0.6294
DAPT g2000 (dag_*)	0.8790	0.7790	0.5978	0.2812	0.2677	0.8118	0.5147
DAPT g5000 (da_*)	0.8836	0.7555	0.5926	0.2589	0.2424	0.7235	0.5000

Table 4: Candidate accuracies from the archived JSON reports. Elite sample sizes are 653/1,959/1,959 for choice/noul/score; public-benchmark sizes match Table 3. The frozen c_* heads are retained for release. Head optimization differs between c_* and the other rows, and bw1_sr is an early-training, shorter-context control. Consequently, this table does not isolate adaptation, model size, or training duration as a cause of the differences.

arm	elite (complete evidence)			GPQA		SciFact	
	choice	noul	score	main	diamond	noul	choice
frozen (fr)	0.970±.017	0.925±.006	0.859±.075	0.295±.014	0.307±.012	0.483±.023	0.479±.018
adapted g5000 (da)	0.961±.011	0.913±.007	0.779±.109	0.288±.018	0.297±.011	0.709±.020	0.444±.062

Table 5: Accuracy (mean $\pm$ sd over seeds 7331–7333) of the matched frozen/adapted arms on `systemone-v2` (identical head recipe, splits, and evaluation per seed). Elite n : 448/1,200/1,218; GPQA main/diamond: 441/193; SciFact: 332 per task. Dev-fitted temperatures for seed 7331: choice 3.39/2.17, noul 1.49/2.78, score 1.27/1.76 (fr/da). Adaptation robustly improves out-of-distribution verification (SciFact noul +22.6 points, consistent across seeds, from below the 0.60 majority baseline) at a noisy cost to in-distribution ordinal scoring. This isolates the backbone effect the candidate screen could not, and reverses the sign of the SciFact result observed in the unmatched comparison.

drops seed-7331 accuracy to 0.7521 (argmax MAE 0.2479 versus 0.1043) with worse calibration—though the three-seed spread (± 0.11) shows this gap fluctuates across runs. The matched design therefore rejects a uniform verdict in either direction: adaptation effects are task-dependent, not monotone, and the verification gain is the most robust difference observed.

4.6 Evidence controls expose option-side leakage

High accuracy alone does not prove that decisions condition on the supplied evidence. We therefore evaluate *evidence controls*: empty (the evidence span removed) and shuffle (the span replaced by a donor passage from a different source group in the same split). Controls keep the original question and proposed answer and are scored for *agreement with the reference answer*—not relabelled, since altered evidence has no valid ground truth. Agreement far above chance under destroyed evidence means the model decides without reading the evidence, whether through parametric recall or option-side cues.

arm	kind	original acc	empty agree	shuffle agree
fr	choice	0.9799	0.8750	0.8725
fr	noul	0.9308	0.8100	0.6693
fr	score	0.8957	0.5747	0.3848
da	choice	0.9598	0.8058	0.7852
da	noul	0.9200	0.8067	0.6667
da	score	0.7521	0.5550	0.4300

Table 6: Evidence controls on the elite eval split (n per cell 447–1,218; shuffle excludes 101 donor-overflow rows). Chance is $\sim 0.25/0.60/0.34$ for choice/noul/score (noul majority prior). Controls are scored as agreement with the reference answer under altered evidence, `label_status=evidence_control`.

The controls qualify the headline accuracies. Choice agreement under empty evidence is 0.87 (frozen): the task is largely solvable without the passage, because gold answers are teacher-written text that matches question phrasing while distractors are numeric perturbations or topically foreign answers. Noul falls to its majority prior under shuffled evidence, as it should. Score behaves as designed—agreement collapses toward chance under wrong evidence—and is the only task whose accuracy directly reflects evidence use. Both arms rely on option-side leakage similarly; adaptation does not fix it. The matched elite numbers should therefore be read as upper bounds on evidence-grounded accuracy, and the battery needs distractors that are indistinguishable without the passage before choice accuracy can be claimed as grounded.

Stricter distractors reduce but do not remove leakage. Rebuilding the choice battery with a strict pool (`--hard-choice`: same-passage swaps and numeric perturbations only, no cross-passage fill; 733 records honestly excluded as untestable, $n = 303$ eval) changes accuracy to 0.914 (fr) / 0.868 (da) and empty-evidence agreement to 0.795 / 0.766. The residual shows that same-passage swaps—answers to *other* questions—remain separable by question–answer coherence alone; fully grounded choice requires confounders to the *same* question. Under shuffled evidence the adapted arm agrees less than the frozen arm (0.722 versus 0.825), consistent with better evidence filtering, matching its SciFact gain.

5 Discussion

What a decision layer buys. A decision layer exposes probabilities without text generation or output parsing. On the elite battery, the frozen heads have ECE 0.0253–0.1221 and choice automation@5% of 0.8070. This retrospective operating curve can inform routing to a slower component, but a deployment threshold must be validated independently under its own data distribution.

Backbone and head must be evaluated together. Readout architecture changes accuracy even with a fixed backbone, while candidate performance also varies with the backbone, context window, and training schedule. The early `bw1_sr` checkpoint cannot establish a size-dependent knowledge ceiling. Likewise, proximity to a published GPQA score under another inference protocol would not prove that our head extracts all available knowledge.

Candidate selection is not a causal adaptation study. Table 4 supports retaining the frozen heads for this release: the adapted candidates do not satisfy the transfer gate despite some improved elite metrics. It does not establish that domain adaptation necessarily sacrifices generalization, nor that a domain-native backbone would avoid that trade-off. The matched comparison in Table 5 holds head optimization fixed on `systemone-v2` and shows the causal effect is task-dependent: adaptation transfers to external verification while degrading ordinal scoring. A fuller study should still repeat seeds, restore optimizer state across interruptions, and reserve untouched evaluation data after selection.

Relation to the ecosystem. Concurrent System-One implementations motivate typed decision interfaces. Our implementation explores a masked-diffusion backbone and deterministic option ordering. A numerical ranking against other systems would require matched dataset conversions, splits, and calibration protocols; the empirical tables here report only our own evaluated candidates.

6 Limitations

Backbone pretraining may include benchmark-adjacent text. Heads are trained on one decision distribution and on at most four options; Banking77 performance is poor. Score remains weaker than choice on the elite battery. Results are single runs, without multi-seed uncertainty estimates. Candidate comparisons differ in optimization, and the small backbone is only an early-training control. External benchmarks informed release selection, so they are not an untouched final test set. We report archived evaluation outputs and their hashes, not a new evaluation of the full backbone during release packaging.

References

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